

Francisco Togni

Electrical Engineer | FPGA · DSP · Embedded R&D

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Summary

Electrical Engineer with a specialization in **Industrial Electronic Systems**, focused on FPGA, DSP, and embedded systems for real-time applications. Experience in the design and validation of low-latency digital and analog processing architectures, industrial instrumentation, and electronic control systems. C, Verilog, HDL/RTL, ARM Cortex-M, TI C2000, Xilinx. Advanced English (C1).

Experience

- **Research Collaborator**, *G.E.A. (Applied Electronics Group), National University of Río Cuarto* 2024 – 2026
R&D and FPGA implementation of low-latency strategies for open-switch fault detection in power semiconductor devices used in electric drives.
 - Designed and implemented HDL/RTL architecture for real-time fault detection in inverter power switches, achieving an average diagnosis time of 10% of the electrical period while using 5% of the FPGA's resources at a base clock frequency of 100 MHz.
 - Published in *IEEE Xplore* and presented at the *8th Argentine Conference on Electronics (CAE)*, UNT, 2026.
- **Service and Network Technician**, *Cablevisión – Fibertel (Telecom)* 2013 – 2019
Diagnosis and resolution of critical faults in HFC (Hybrid Fiber-Coaxial) infrastructure; advanced technical support and network operations.
- **Self-employed Electronics Engineer**, *Electronic Solutions* 2011 – 2012
Project management and direct client support; R&D of custom electronic solutions for industrial and commercial clients.

Selected Projects

- **Amplitude-Dependent Nonlinear System Identification via Volterra Kernels and PCA** 2026
Development of a method for identifying and synthesizing amplitude-dependent nonlinear systems.
 - Identification of nonlinear systems using Volterra kernels and PCA from impulse responses at varying excitation levels.
 - FPGA architecture: parallelized, pipelined single-convolution engine for real-time synthesis.
- **Professional Ball Launcher — Trinquete (client product, IP retained)** 2026
Design and development of an end-to-end mechatronic system for professional training.
 - Distributed dual-MCU architecture (TI TMS320F280049 engine master + STM32 supervisor, linked via Modbus RTU) with multi-axis motion control (speed, spin, tilt, azimuth), sensor telemetry (current sensors, encoders), and a local HMI for operational control.
 - Full mechanical, electrical (PCB), and 3D (SolidWorks) design.
- **Adaptive Decimation for Real-Time Pitch Shifting** 2023
Real-time pitch shifter correlated to musical intervals (just and equal temperament scales).
 - ARM Cortex-M4F (STM32 Black Pill), C firmware, 44.1 kHz sampling, 2 ms latency.
- **Digital Audio Equalizer on FPGA** 2023
160-tap FIR filter bank with 4-level PWM-based output DAC topology.
 - 44.1 kHz sampling, low FPGA resource usage.
 - Published in *Proc. 28th AADECA Congress*, University of Palermo, Buenos Aires, 2023.
- **Multi-Channel Temperature Control for Industrial Lab Flasks** 2026
Six-channel simultaneous temperature control system for laboratory equipment.
 - SSR-based burst-fire control strategy with 500 ms window and NTC sensing, implemented in C firmware on ARM Cortex-M.
 - Local UI with 7-segment display and rotary encoder, designed for continuous industrial operation.

Publications and Presentations

· F. Togni <i>et al.</i> , “ Low-Latency Architecture for Open-Circuit Fault Diagnosis in Electric Drive Inverter Power Switches ↗ ,” in <i>Proc. 8th Argentine Conference on Electronics (CAE)</i> , UNT, 2026.	2026
· F. Togni <i>et al.</i> , “ Digital Audio Equalizer Implementation using an FPGA with FIR Filter Bank Structure and 4-Level PWM-Based Output DAC Topology ↗ ,” in <i>Proc. 28th AADECA Congress</i> , University of Palermo, Buenos Aires, 2023.	2023

Skills

· Programming: C, Verilog, Python
· FPGA / HDL: Xilinx Vivado, RTL synthesis & optimization, timing analysis, HDL/IP integration, FPGA-based DSP
· Embedded: ARM Cortex-M, TI C2000, STM32, PIC
· DSP: FIR filter banks, Volterra kernels, PCA, real-time convolution, PWM-based DAC
· Hardware: ADC/DAC, DMA, SRAM, analog signal conditioning, motor control, power electronics.
· Communications: I ² C, I ² S, SPI, UART/RS-485, USB; Modbus RTU/TCP, USB Device/Host
· Tools: Altium Designer, SolidWorks, MATLAB/Simulink
· Industrial: PLC/HMI (Siemens, Kinco); burst-fire control, multi-channel process control

Education

· Electrical Engineer , <i>National University of Río Cuarto</i> , Río Cuarto, Argentina.	2019 – 2026
– Specialization in Industrial Electronic Systems	
· Higher Technical Degree in Electromechanics , <i>Technological Institute of Río Cuarto</i> , Argentina.	2009 – 2011
· Mechanical Engineering (incomplete) , <i>National University of Río Cuarto</i> , Río Cuarto, Argentina.	2006 – 2008

Courses and Certifications

· <i>From Algorithm to Hardware: Machine Learning in Embedded Systems</i> , UNMdP.	2025
· <i>DSP for Power Electronics Control</i> , U.N.R.C.	2024
· <i>English C1 Advanced</i> , EF SET Certificate ↗ .	2024
· <i>Advanced Programming in C for PIC Microcontrollers</i> , Microchip Technology Inc.	2013

Languages

Spanish (native) · English (C1 Advanced)
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